

HW0414_314707031

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Exercise 8.16: Vacation Travel Model

We consider the regression model

$$MILES_i = \beta_1 + \beta_2 INCOME_i + \beta_3 AGE_i + \beta_4 KIDS_i + e_i,$$

where **MILES** is miles driven per year, **INCOME** is measured in thousand-dollar units, **AGE** is the average age of adult household members, and **KIDS** is the number of children in the household.

The parameter of main interest is β_4 , which measures the effect of one additional child on miles traveled, holding income and age constant.

Load packages and data

```
# Packages used in this homework
library(sandwich)
library(lmtest)

# Put the vacation data file in the same folder as this Rmd file.
# This chunk tries several common file formats automatically.

vacation <- read.csv("vacation.csv")

# Standardize variable names to upper case, so the code works even if the data uses lower-case names.
names(vacation) <- toupper(names(vacation))

# Keep the variables needed in this exercise and remove incomplete observations.
vacation <- na.omit(vacation[, c("MILES", "INCOME", "AGE", "KIDS")])

str(vacation)

## 'data.frame':    200 obs. of  4 variables:
## $ MILES : int  902 491 1841 406 0 1899 724 1404 1367 560 ...
## $ INCOME: int   41 31 87 54 77 70 43 40 69 38 ...
## $ AGE   : int   26 38 40 48 43 55 24 39 55 29 ...
## $ KIDS  : int    0 3 2 4 4 2 0 0 2 2 ...
```

```
summary(vacation)
```

```
##           MILES           INCOME           AGE           KIDS
## Min.      : 0.0    Min.      : 19.00   Min.      :23.00   Min.      :0.000
## 1st Qu.: 667.5    1st Qu.: 51.00   1st Qu.:35.00   1st Qu.:0.000
## Median :1003.0    Median : 62.00   Median :43.00   Median :2.000
## Mean     :1054.2    Mean     : 63.92   Mean     :42.67   Mean     :1.635
## 3rd Qu.:1445.5    3rd Qu.: 78.00   3rd Qu.:50.25   3rd Qu.:3.000
## Max.     :2609.0    Max.     :119.00   Max.     :59.00   Max.     :4.000
```

(a) OLS estimation and 95% confidence interval for KIDS

```
ols_model <- lm(MILES ~ INCOME + AGE + KIDS, data = vacation)
summary(ols_model)
```

```
##
## Call:
## lm(formula = MILES ~ INCOME + AGE + KIDS, data = vacation)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1198.14  -295.31    17.98   287.54  1549.41
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -391.548    169.775  -2.306  0.0221 *
## INCOME         14.201     1.800   7.889 2.10e-13 ***
## AGE           15.741     3.757   4.189 4.23e-05 ***
## KIDS          -81.826    27.130  -3.016  0.0029 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 452.3 on 196 degrees of freedom
## Multiple R-squared:  0.3406, Adjusted R-squared:  0.3305
## F-statistic: 33.75 on 3 and 196 DF, p-value: < 2.2e-16
```

```
# 95% conventional OLS confidence interval for beta_KIDS
confint(ols_model, "KIDS", level = 0.95)
```

```
##           2.5 %    97.5 %
## KIDS -135.3298 -28.32302
```

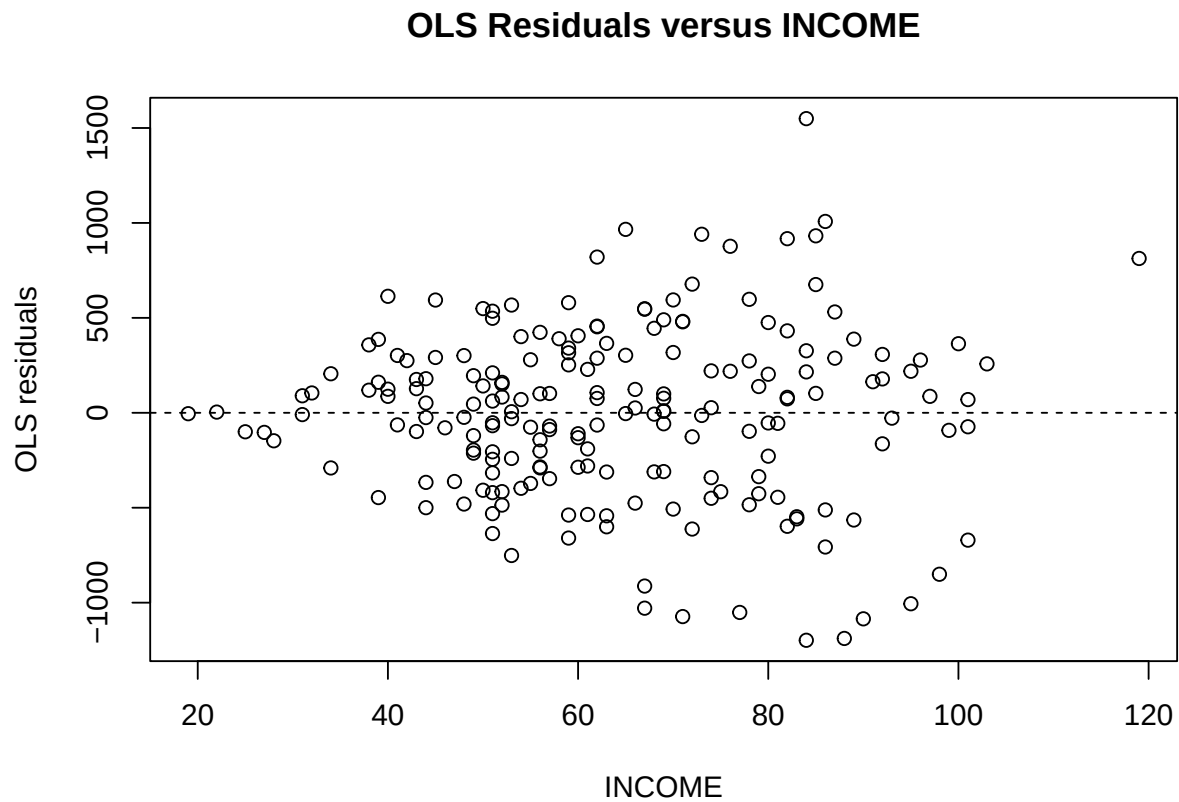
Explanation.

The coefficient on KIDS estimates the marginal effect of one more child on annual miles traveled, holding INCOME and AGE constant. The 95% confidence interval gives a range of plausible values for this partial effect. If the interval contains zero, then the estimated effect is not statistically different from zero at the 5% significance level under the usual homoskedastic OLS assumptions. Humanity has once again converted family vacations into a regression coefficient, which is somehow both tragic and useful.

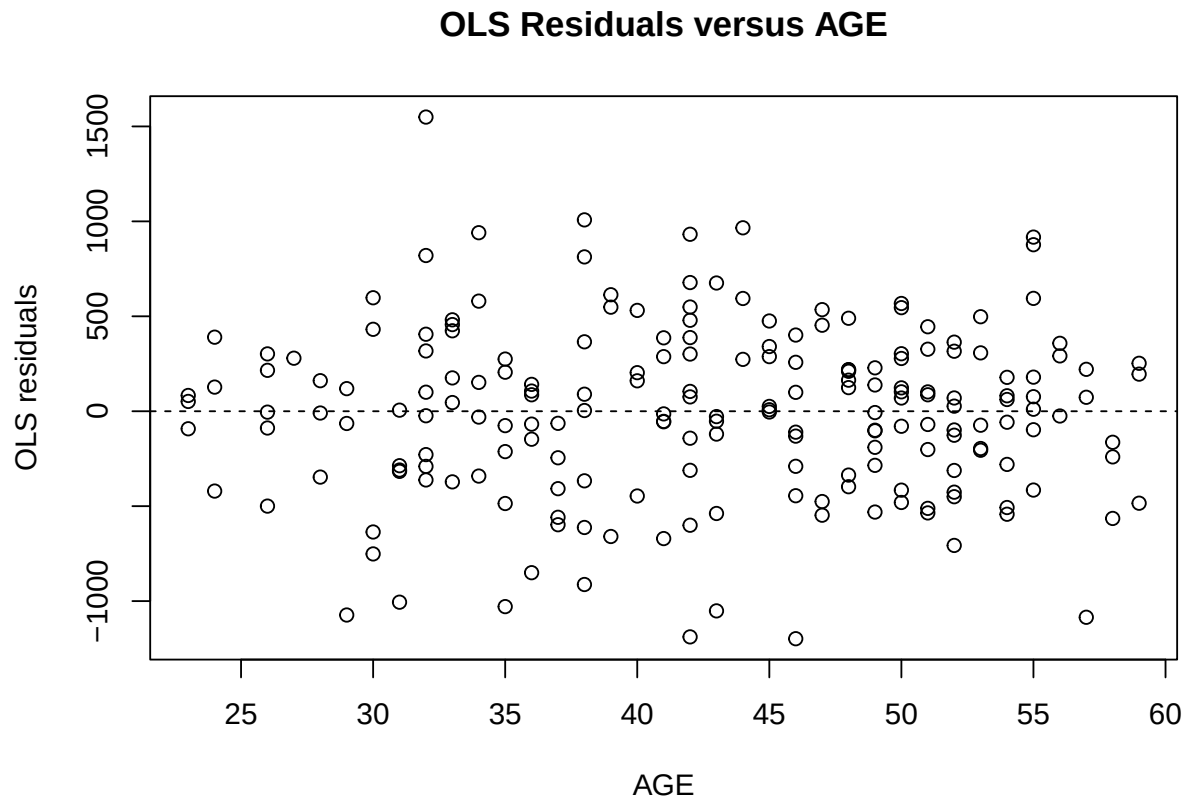
(b) Residual plots against INCOME and AGE

```
vacation$resid_ols <- resid(ols_model)

plot(vacation$INCOME, vacation$resid_ols,
     xlab = "INCOME",
     ylab = "OLS residuals",
     main = "OLS Residuals versus INCOME")
abline(h = 0, lty = 2)
```



```
plot(vacation$AGE, vacation$resid_ols,
     xlab = "AGE",
     ylab = "OLS residuals",
     main = "OLS Residuals versus AGE")
abline(h = 0, lty = 2)
```



Explanation.

These residual plots are used to visually check whether the error variance appears constant. If the residuals spread out more as INCOME or AGE increases, this suggests heteroskedasticity. If the spread is roughly constant around zero, there is less visual evidence of heteroskedasticity. The key pattern to look for is a funnel shape or systematic change in residual dispersion, not whether individual residuals look emotionally unstable, even though they usually do.

(c) Goldfeld–Quandt test sorted by INCOME

```
# Sort observations by increasing income
vacation_sorted <- vacation[order(vacation$INCOME), ]

# Use the first 90 and last 90 observations
low_income <- vacation_sorted[1:90, ]
high_income <- vacation_sorted[(nrow(vacation_sorted) - 89):nrow(vacation_sorted), ]

ols_low <- lm(MILES ~ INCOME + AGE + KIDS, data = low_income)
ols_high <- lm(MILES ~ INCOME + AGE + KIDS, data = high_income)

SSE_low <- sum(resid(ols_low)^2)
SSE_high <- sum(resid(ols_high)^2)

k <- length(coef(ols_model))
```

```

df_low <- nrow(low_income) - k
df_high <- nrow(high_income) - k

# Goldfeld-Quandt statistic for the alternative that variance is larger at high income
GQ_stat <- (SSE_high / df_high) / (SSE_low / df_low)
GQ_crit <- qf(0.95, df_high, df_low)
GQ_pval <- 1 - pf(GQ_stat, df_high, df_low)

GQ_result <- data.frame(
  SSE_low = SSE_low,
  SSE_high = SSE_high,
  df_low = df_low,
  df_high = df_high,
  GQ_statistic = GQ_stat,
  critical_value_5_percent = GQ_crit,
  p_value = GQ_pval
)

GQ_result

##   SSE_low SSE_high df_low df_high GQ_statistic critical_value_5_percent
## 1 8750040 27160654    86    86      3.104061              1.428617
##      p_value
## 1 1.64001e-07

```

Explanation.

The Goldfeld-Quandt test checks whether the error variance changes with income. The data are sorted by `INCOME`; the middle observations are omitted by using only the first 90 and last 90 observations. The hypotheses are

$$H_0 : \sigma_{low}^2 = \sigma_{high}^2$$

against

$$H_1 : \sigma_{high}^2 > \sigma_{low}^2.$$

The test statistic is

$$F = \frac{SSE_{high}/df_{high}}{SSE_{low}/df_{low}}.$$

At the 5% significance level, reject H_0 if the test statistic is larger than the 95th percentile of the $F(df_{high}, df_{low})$ distribution, or equivalently if the p-value is less than 0.05. Rejection means there is evidence that the variance of the errors is larger for high-income households. Apparently richer households not only travel differently, their regression errors also demand premium seating.

(d) OLS with heteroskedasticity-robust standard errors

```

# HC1 is a common finite-sample robust variance estimator
ols_robust_vcov <- vcovHC(ols_model, type = "HC1")
coeftest(ols_model, vcov = ols_robust_vcov)

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -391.5480   142.6548 -2.7447 0.0066190 **
## INCOME      14.2013     1.9389   7.3246 6.083e-12 ***
## AGE         15.7409     3.9657   3.9692 0.0001011 ***
## KIDS        -81.8264    29.1544 -2.8067 0.0055112 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# 95% robust confidence interval for beta_KIDS
beta_kids_ols <- coef(ols_model)["KIDS"]
se_kids_robust <- sqrt(diag(ols_robust_vcov))["KIDS"]
t_crit_ols <- qt(0.975, df = df.residual(ols_model))

robust_CI_kids <- beta_kids_ols + c(-1, 1) * t_crit_ols * se_kids_robust
robust_CI_kids

## [1] -139.32297 -24.32986

```

Explanation.

The coefficient estimate for KIDS is the same as in part (a), because the model is still estimated by OLS. What changes is the standard error. Robust standard errors allow for heteroskedasticity, so the resulting confidence interval is more reliable if the error variance is not constant. Compare this interval with the interval in part (a). If the robust interval is wider, the conventional OLS standard error was likely too small. If it is narrower, the conventional standard error was larger than the robust one. Either way, the coefficient itself does not change; only our uncertainty changes, because uncertainty enjoys making students suffer.

(e) GLS assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$

If

$$\sigma_i^2 = \sigma^2 INCOME_i^2,$$

then the error variance is proportional to `INCOME` squared. The appropriate GLS weights are proportional to the inverse of the error variance:

$$w_i = \frac{1}{INCOME_i^2}.$$

Thus, the model can be estimated by weighted least squares.

```

gls_model <- lm(MILES ~ INCOME + AGE + KIDS,
               data = vacation,
               weights = 1 / INCOME^2)

summary(gls_model)

##
## Call:
## lm(formula = MILES ~ INCOME + AGE + KIDS, data = vacation, weights = 1/INCOME^2)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1907  -4.9555   0.2488   4.3832  18.5462
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -424.996    121.444  -3.500 0.000577 ***
## INCOME         13.947     1.481   9.420 < 2e-16 ***
## AGE           16.717     3.025   5.527 1.03e-07 ***
## KIDS          -76.806    21.848  -3.515 0.000545 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.765 on 196 degrees of freedom
## Multiple R-squared:  0.4573, Adjusted R-squared:  0.449
## F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16

# Conventional GLS confidence interval for beta_KIDS
confint(gls_model, "KIDS", level = 0.95)

##              2.5 %      97.5 %
## KIDS -119.8945 -33.71808

# Robust standard errors for the weighted least squares / GLS model
gls_robust_vcov <- vcovHC(gls_model, type = "HC1")
coeftest(gls_model, vcov = gls_robust_vcov)

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -424.9962    95.8035  -4.4361 1.526e-05 ***
## INCOME        13.9473     1.3470 10.3545 < 2.2e-16 ***
## AGE          16.7175     2.7974  5.9761 1.061e-08 ***
## KIDS         -76.8063    22.6186  -3.3957 0.0008286 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# 95% robust GLS confidence interval for beta_KIDS
beta_kids_gls <- coef(gls_model)["KIDS"]
se_kids_gls_robust <- sqrt(diag(gls_robust_vcov))["KIDS"]

```

```
t_crit_gls <- qt(0.975, df = df.residual(gls_model))

glb_robust_CI_kids <- beta_kids_gls + c(-1, 1) * t_crit_gls * se_kids_gls_robust
glb_robust_CI_kids

## [1] -121.41339 -32.19919
```

Explanation.

The GLS model uses weights equal to $1/INCOME_i^2$, because observations with larger income are assumed to have larger error variance. Under the stated variance assumption, GLS is more efficient than OLS. The conventional GLS confidence interval is valid if the variance specification $\sigma_i^2 = \sigma^2 INCOME_i^2$ is correct. The robust GLS confidence interval is safer because it allows the variance model to be imperfect.

Compared with part (a), GLS may produce a different coefficient estimate because it gives less weight to high-income observations. Compared with part (d), robust GLS changes both the coefficient estimate and the standard error, while robust OLS only changes the standard error. The preferred interval depends on whether the assumed variance pattern is credible. If the variance truly grows with income squared, GLS should be more efficient. If that assumption is wrong, robust OLS or robust GLS is less fragile than pretending the world is tidy, which it absolutely is not.

Summary table for the KIDS effect

```
# Conventional OLS CI
ols_CI <- confint(ols_model, "KIDS", level = 0.95)

# Conventional GLS CI
glb_CI <- confint(gls_model, "KIDS", level = 0.95)

summary_table <- data.frame(
  Method = c("OLS conventional", "OLS robust", "GLS conventional", "GLS robust"),
  Estimate = c(beta_kids_ols, beta_kids_ols, beta_kids_gls, beta_kids_gls),
  Lower_95 = c(ols_CI[1], robust_CI_kids[1], glb_CI[1], glb_robust_CI_kids[1]),
  Upper_95 = c(ols_CI[2], robust_CI_kids[2], glb_CI[2], glb_robust_CI_kids[2])
)

summary_table

##           Method Estimate Lower_95 Upper_95
## 1 OLS conventional -81.82642 -135.3298 -28.32302
## 2 OLS robust -81.82642 -139.3230 -24.32986
## 3 GLS conventional -76.80629 -119.8945 -33.71808
## 4 GLS robust -76.80629 -121.4134 -32.19919
```

Final interpretation.

This table compares the estimated effect of one additional child across OLS, robust OLS, GLS, and robust GLS. The comparison should focus on two things: whether the point estimate changes, and whether the confidence interval becomes wider or narrower. OLS and robust OLS have the same point estimate but different standard errors. GLS and robust GLS may have a different point estimate because the observations are weighted by income. If the confidence intervals all lead to the same qualitative conclusion, then the result is fairly robust. If the conclusion changes across methods, then the inference depends heavily on how heteroskedasticity is handled.